

Task 3 Pen & Paper

Note: the formulations for $P(x, t)$ are equation (25) and equation (30).

For this task I will derive the relationship between the boundary crossings and the function itself, then use that to arrive at a formula for what these boundary crossings should look like.

First units. Let $P(x, t)$ be our function for how many particles cross a point at location x at time t in the rightward direction, per unit time. These units are useful because we will come to see that less particles cross boundaries for smaller time steps.

Next: what must definitely be true about our function P ? Consider a region (a, b) of our domain in x . Particles can only enter and leave at the boundaries a and b , and how many particles cross these boundaries over time is $P(a, t)$ and $P(b, t)$ respectively. Assuming $b > a$, then particles heading right at b are leaving, and particles heading right at a are entering, so the number of particles leaving over time is $P(a, t) - P(b, t)$.

Is there another formula for the same thing? Yes! The change over time in the number of particles is also simply the below:

$$\frac{\partial}{\partial t} \int_a^b n \, dx \quad (1)$$

The integral gets us the particles between a and b , and the derivative gets us the change in that number over time. So we have:

$$P(a, t) - P(b, t) = \frac{\partial}{\partial t} \int_a^b n \, dx \quad (2)$$

$$P(a, t) - P(b, t) = \int_a^b \frac{\partial n}{\partial t} \, dx \quad (3)$$

Recall the heat equation that $n(x, t)$ satisfies:

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2} \quad (4)$$

We can substitute this in:

$$P(a, t) - P(b, t) = \int_a^b D \frac{\partial^2 n}{\partial x^2} \, dx \quad (5)$$

$$(6)$$

Now let's see if we can find a way to get rid of the a's and b's and just get a formulation in x . Note that the left hand side looks like the fundamental

theorem of calculus can be applied:

$$\int_b^a \frac{\partial P}{\partial x} dx = \int_a^b D \frac{\partial^2 n}{\partial x^2} dx \quad (7)$$

$$\int_a^b -\frac{\partial P}{\partial x} dx = \int_a^b D \frac{\partial^2 n}{\partial x^2} dx \quad (8)$$

$$(9)$$

And we get two equal integrals whose bounds are the same. However, the bounds a and b are arbitrary. This means that the integrands must be equal:

$$-\frac{\partial P}{\partial x} = D \frac{\partial^2 n}{\partial x^2} \quad (10)$$

Now just integrate indefinitely in x and we get:

$$P(x, t) = -D \frac{\partial n}{\partial x} + C(t) \quad (11)$$

Note that our constant of integration becomes time dependent since we only know it's a constant in x .

We can get rid of this constant by imposing a constraint that we expect the real world particles to follow. These are:

- **Locality**, or that in a very small time step, the way that particles cross a given point x at some time t can only depend on the particles in the immediate surroundings of that point. This means that our constant function $C(t)$ doesn't depend on the global distribution of particles $n(x, t)$ and is the same for all distributions.
- **Symmetry** under a flip in the x direction. The particles shouldn't have a preference for one direction over another.

Such a transformation looks like the below:

$$\hat{n}(x, t) = n(-x, t) \quad (12)$$

$$\hat{P}(x, t) = -P(-x, t) \quad (13)$$

For the second one, the negative arises because all particles going right originally are going left after the transformation. This implies:

$$\hat{P}(x, t) = -D \frac{\partial}{\partial x} \hat{n} + C(t) = -P(-x, t) \quad (14)$$

$$-P(-x, t) = -D \frac{\partial}{\partial x} (n(-x, t)) + C(t) \quad (15)$$

We can apply the chain rule to the derivative in (14):

$$-P(-x, t) = D \frac{\partial n}{\partial x}(-x, t) + C(t) \quad (16)$$

And apply (10) on the left side:

$$D \frac{\partial n}{\partial x}(-x, t) - C(t) = D \frac{\partial n}{\partial x}(-x, t) + C(t) \quad (17)$$

$$-C(t) = C(t) \quad (18)$$

$$C(t) = 0 \quad (19)$$

And we have gotten rid of the constant. So the relationship we are left with is:

$$P(x, t) = -D \frac{\partial n}{\partial x} \quad (20)$$

We can finally use this to get our answers. The first formulation from task 2 was:

$$n(x, t) = \frac{N}{2\sqrt{\pi Dt}} \sum_m^{m \in \mathbb{Z}} e^{-\frac{(x-x_0+mL)^2}{4Dt}} \quad (21)$$

Applying the formula:

$$P(x, t) = -D \frac{\partial}{\partial x} \frac{N}{2\sqrt{\pi Dt}} \sum_m^{m \in \mathbb{Z}} e^{-\frac{(x-x_0+mL)^2}{4Dt}} \quad (22)$$

$$P(x, t) = -D \frac{N}{2\sqrt{\pi Dt}} \sum_m^{m \in \mathbb{Z}} \frac{\partial}{\partial x} e^{-\frac{(x-x_0+mL)^2}{4Dt}} \quad (23)$$

$$P(x, t) = -D \frac{N}{2\sqrt{\pi Dt}} \sum_m^{m \in \mathbb{Z}} -\frac{2(x-x_0+mL)}{4Dt} e^{-\frac{(x-x_0+mL)^2}{4Dt}} \quad (24)$$

$$P(x, t) = \frac{N}{4\sqrt{\pi Dt^3}} \sum_m^{m \in \mathbb{Z}} (x-x_0+mL) e^{-\frac{(x-x_0+mL)^2}{4Dt}} \quad (25)$$

The second one was:

$$n(x, t) = \frac{N}{L} + \frac{2N}{L} \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \left(\cos \frac{2n\pi}{L} (x-x_0) \right) \quad (26)$$

Applying the rule:

$$P(x, t) = -D \frac{\partial}{\partial x} \frac{2N}{L} \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \left(\cos \frac{2n\pi}{L} (x-x_0) \right) \quad (27)$$

$$P(x, t) = -D \frac{2N}{L} \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \frac{\partial}{\partial x} \left(\cos \frac{2n\pi}{L} (x-x_0) \right) \quad (28)$$

$$P(x, t) = D \frac{2N}{L} \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \frac{2n\pi}{L} \sin \frac{2n\pi}{L} (x-x_0) \quad (29)$$

$$P(x, t) = \frac{4\pi DN}{L^2} \sum_{n=1}^{\infty} n e^{-D(\frac{2n\pi}{L})^2 t} \sin \frac{2n\pi}{L} (x-x_0) \quad (30)$$